

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Understand AI-based code completion tools
- Generate structured programs using AI assistance
- Improve coding speed and productivity
- Validate and refine generated code

Task 1: Class Implementation

Python Code

```
class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

    def display(self):
        print("Name:", self.name)
        print("Marks:", self.marks)

s = Student("Sanju", 90)
s.display()
```

Task 2: Loop Completion

Python Code

```
for i in range(1, 6):  
    print("Number:", i)
```

Task 3: Conditional Statements

Python Code

```
num = int(input("Enter number: "))  
  
if num > 0:  
    print("Positive")  
elif num < 0:  
    print("Negative")  
else:  
    print("Zero")
```

Task 4: Function Completion

Python Code

```
def add(a, b):  
    return a + b  
  
print(add(5, 3))
```

Task 5: Mini Project – Bank System

Python Code

```
class Bank:  
    def __init__(self):  
        self.balance = 0  
  
    def deposit(self, amount):  
        self.balance += amount  
        print("Deposited:", amount)
```

```
def withdraw(self, amount):
    if amount <= self.balance:
        self.balance -= amount
        print("Withdrawn:", amount)
    else:
        print("Insufficient balance")

def show_balance(self):
    print("Balance:", self.balance)

b = Bank()
b.deposit(1000)
b.withdraw(500)
b.show_balance()
```

Conclusion

AI-assisted code completion tools significantly improve coding efficiency.

They help generate boilerplate code quickly, but developers must ensure correctness and optimize logic.